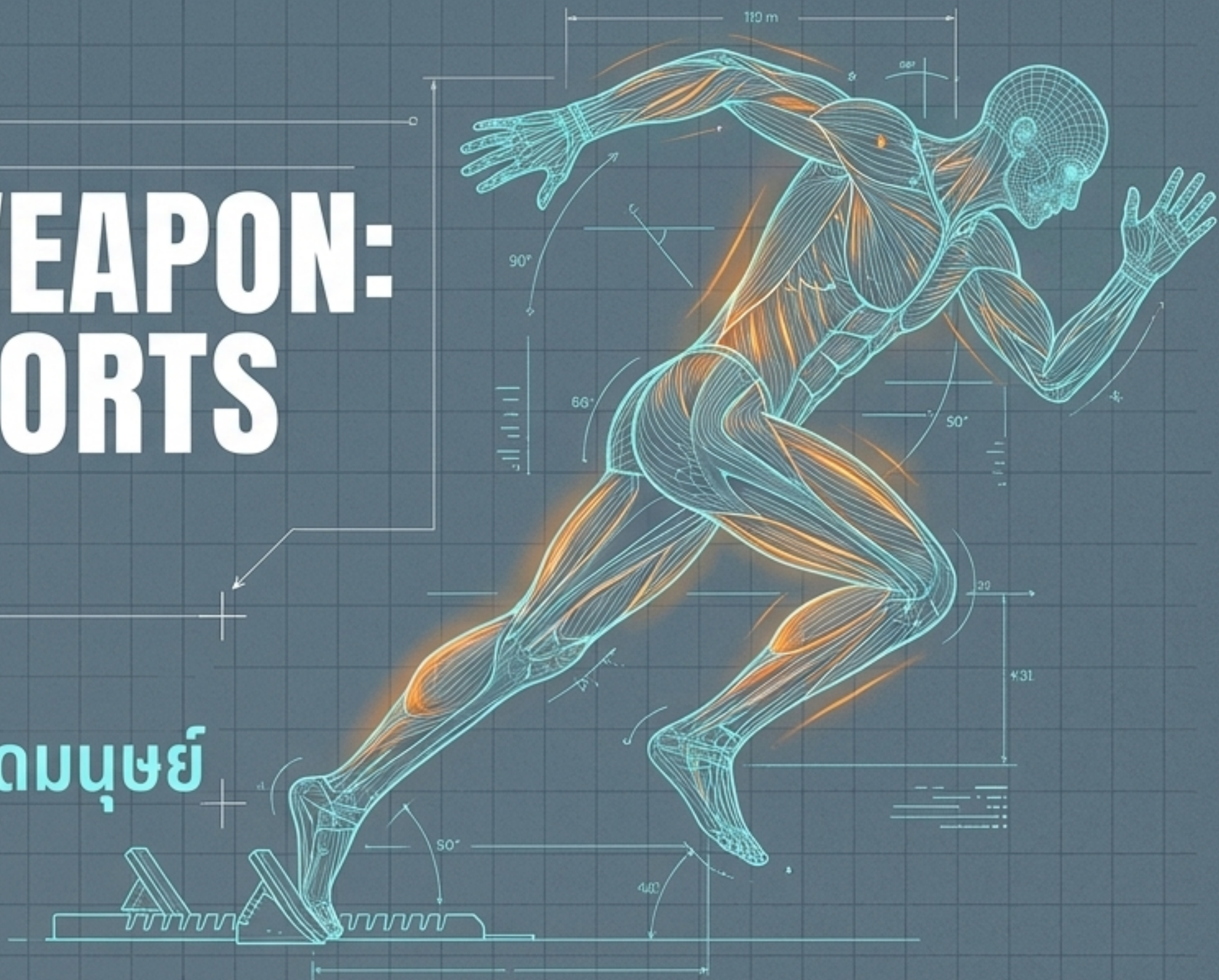


THE SECRET WEAPON: DECODING SPORTS SCIENCE

วิทยาศาสตร์การกีฬา:
ถอดรหัสลับก้าวข้ามขีดจำกัดมนุษย์



Breaking the Impossible

ก้าวข้ามขีดจำกัดของมนุษย์

อดีต: เน้นอึด เน้นแข็งแรง
(The Reality: Athletes relied solely on brutal physical endurance)



The Milestone: Eliud Kipchoge's historic sub-2-hour pace marathon.



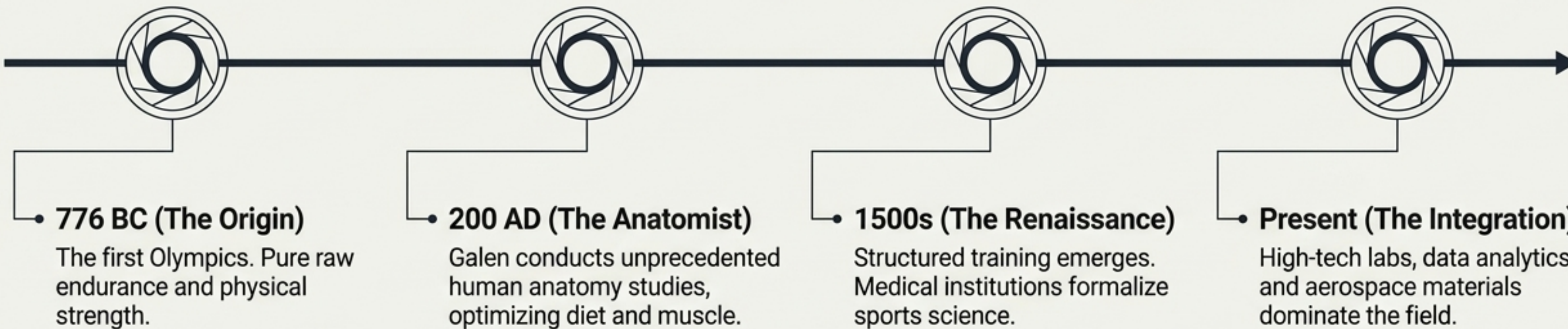
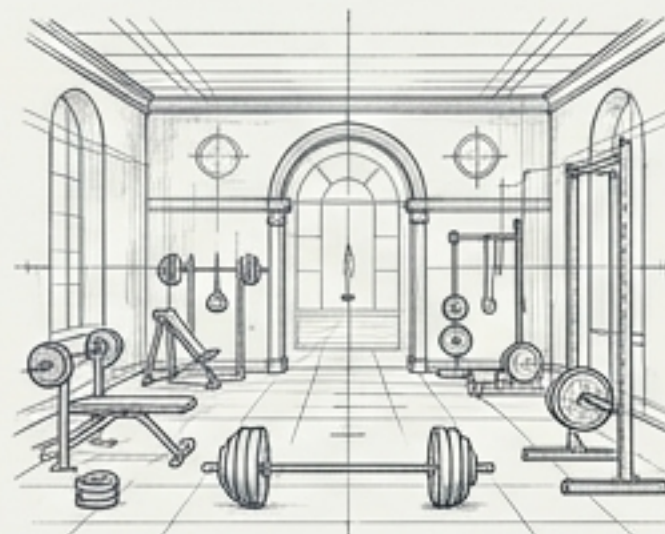
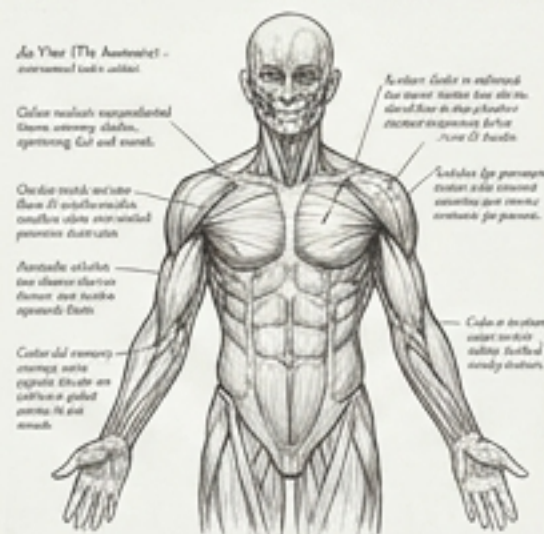
ปัจจุบัน: การผสานกายวิภาค
จิตวิทยา และเทคโนโลยีสุดล้ำ

(The Definition: Sports Science is the multi-disciplinary integration designed to shatter human limits)



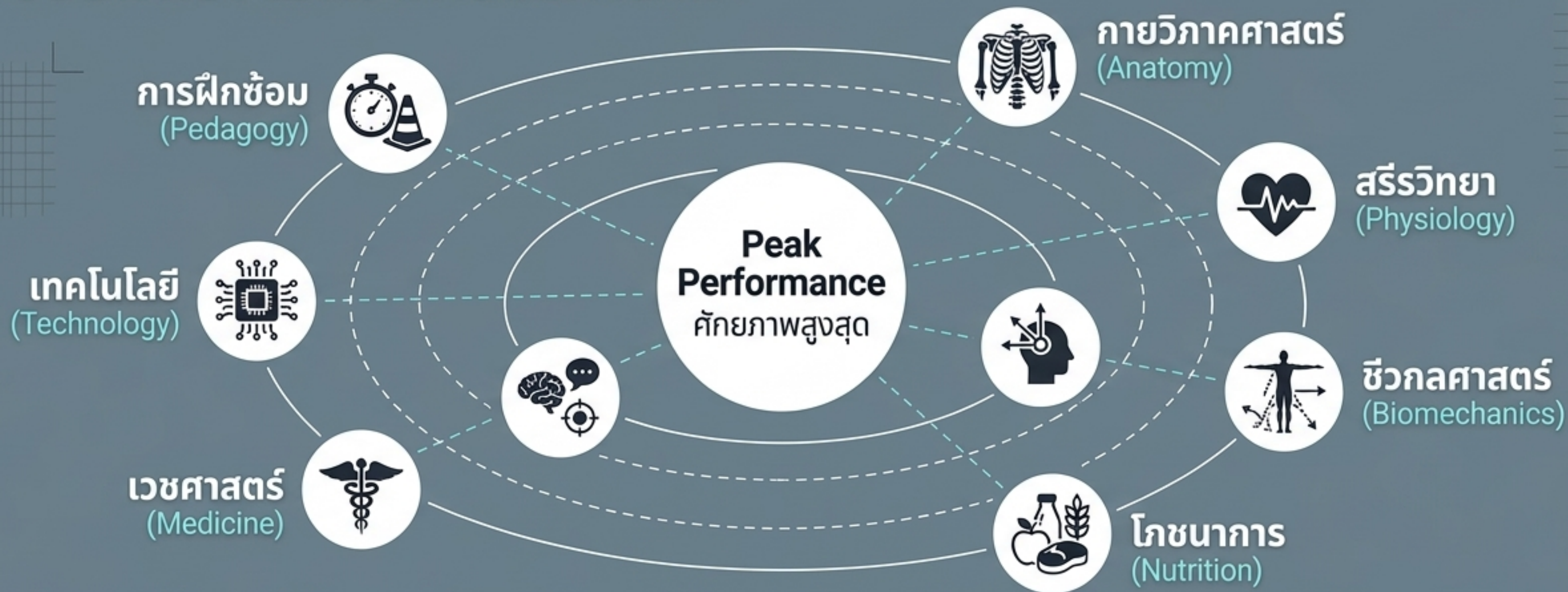
The Evolution of Performance

วิวัฒนาการของวิทยาศาสตร์การกีฬา



The Anatomy of a Champion

ระบบนิเวศของวิทยาศาสตร์การกีฬา

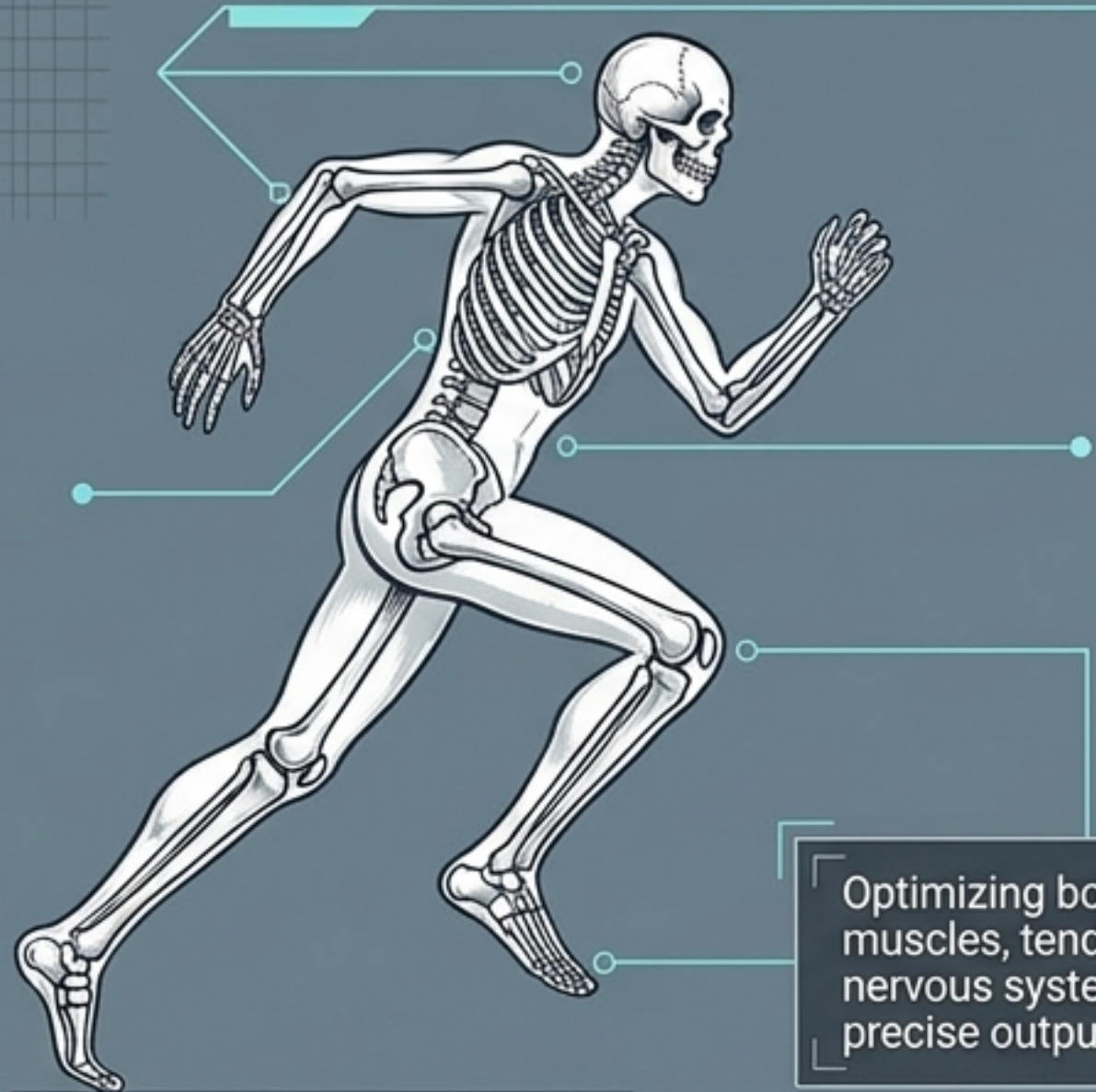


Key Takeaway: It is an applied science—combining these distinct fields into one targeted formula for victory.

The Engine & The Fuel

กายวิภาคและสรีรวิทยา

The Engine (โครงสร้างร่างกาย)



Optimizing bones, muscles, tendons, and the nervous system for precise output.

The Fuel (การทำงานของระบบ)



Energy Systems:

Aerobic vs. Anaerobic, and lactic acid tolerance.

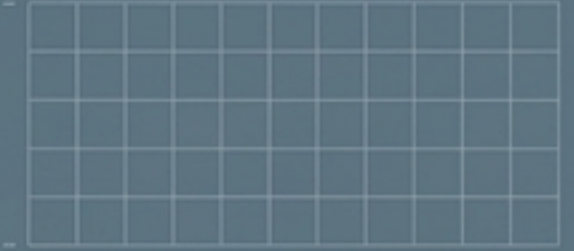
Formula:

$(220 - \text{Age}) * \text{Target \%}$

e.g., A 20-year-old targeting 65% intensity.

The Physics of Winning

ชีวกลศาสตร์ - Biomechanics



The Concept:

Translating muscle and bone movement into pure physical force.



The Application:

Identifying technical flaws, calculating optimal launch angles, and minimizing drag to maximize efficiency.



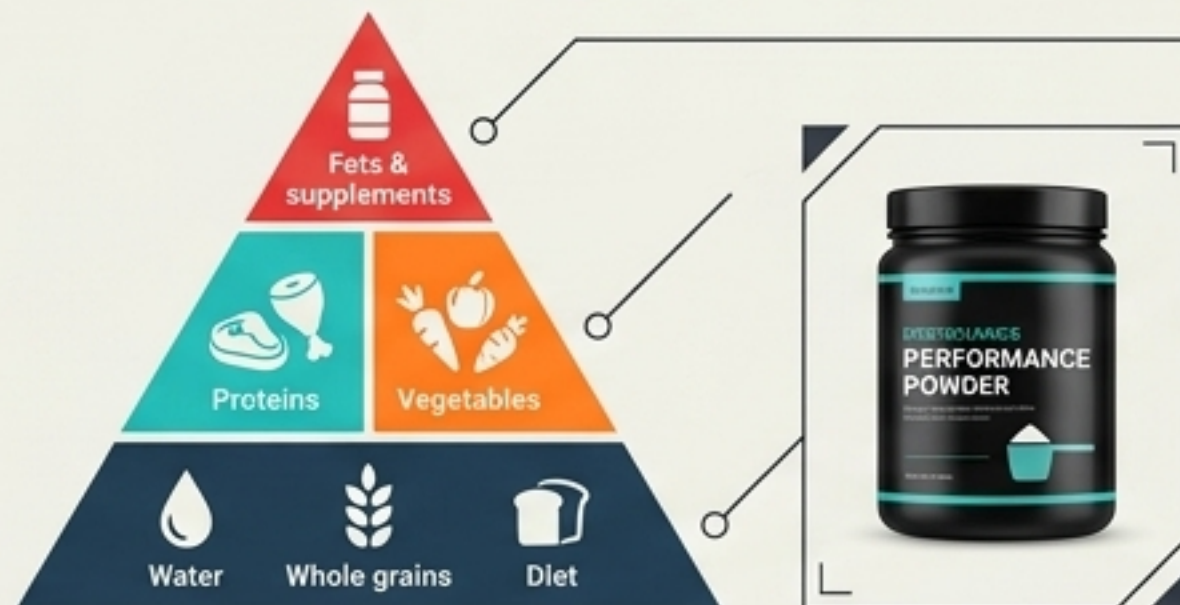
The Mind & The Machine

จิตวิทยาและโภชนาการ



Psychology (จิตวิทยาการกีฬา)

- **Process:** Controlling movement, rapid decision-making under pressure.
- **Goal:** Generating positive drive and reducing performance anxiety.

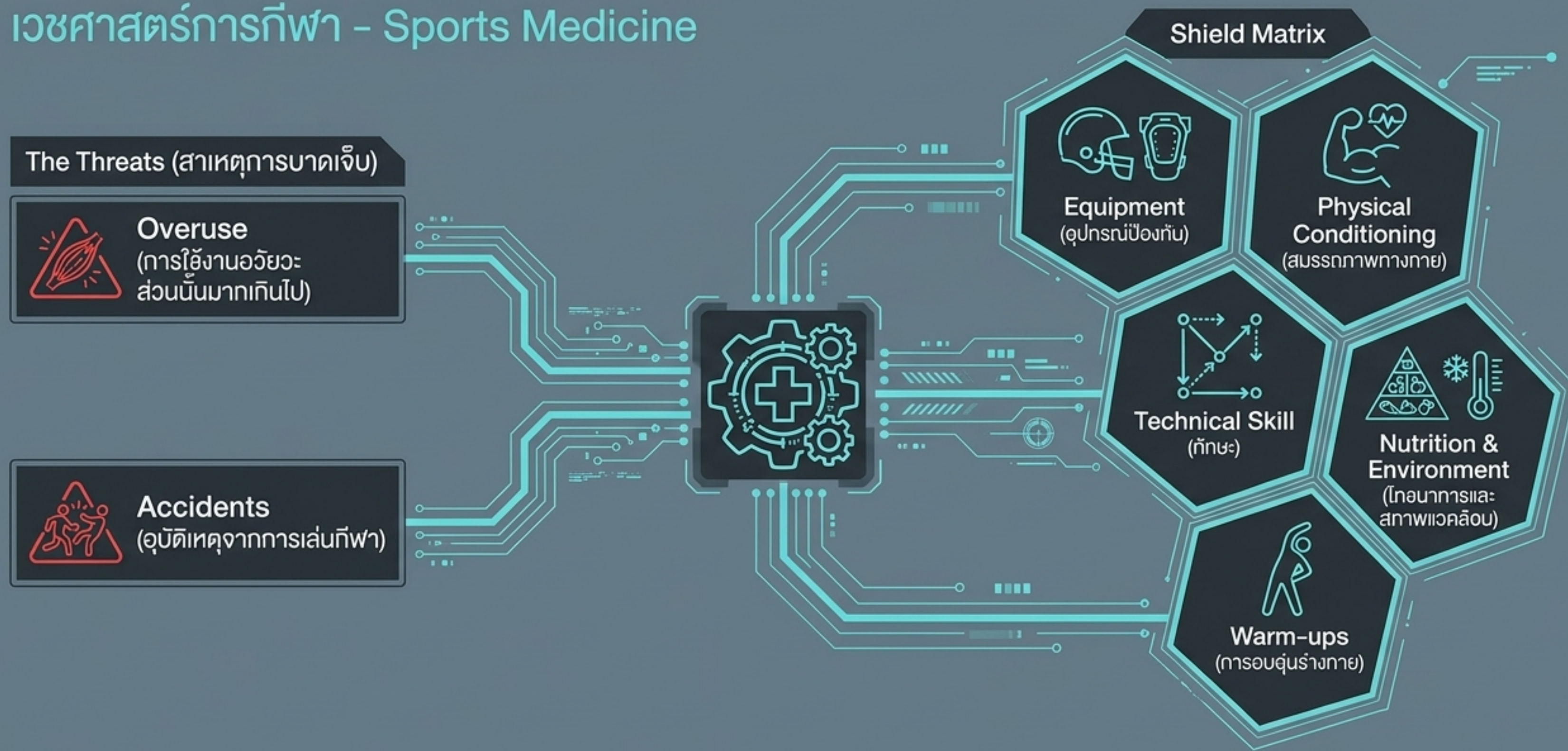


Nutrition (โภชนาการทางการกีฬา)

- **Input:** Targeted energy sources for peak output and rapid cellular repair.
- **Formula:** The ultimate weight-loss & performance stack: Aerobic Exercise + Precise Diet.

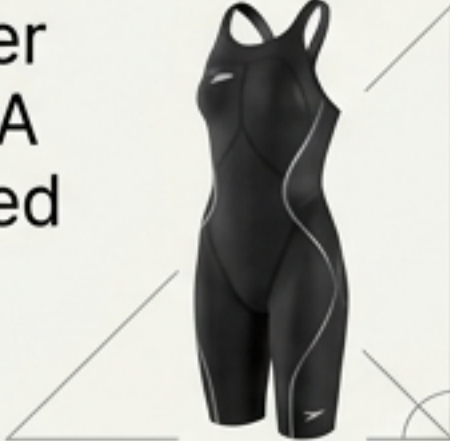
The Maintenance Crew

เวชศาสตร์การกีฬา - Sports Medicine



The Gear Matrix: Banned Tech & Super Shoes

นวัตกรรมเปลี่ยนเกม

Standard Gear	Sports Science Upgrade	The Result
Swimwear (ชุดว่ายน้ำ)		
Two-piece / Standard Polyester. 	Speedo LZR Racer (One-piece, NASA wind-tunnel tested Polyurethane). 	Massive drag reduction. Banned by FINA for providing an unfair advantage.
Footwear (รองเท้าวิ่ง)		
Basic rubber traction, heavy. 	Nike Zoom Superfly. 	Honeycomb traction yields 4x more grip ; ultra-lightweight materials shatter records.

Engineering the Perfect Object

ลูกฟุตบอลสุดไฮเทค: Al Rihla

CRT Core

Central technology that retains air and maintains shape perfectly.



Speed Shell

High-quality polyurethane skin for aerodynamic stability, speed, and passing accuracy.

Eco-Friendly

CFC-free production with minimal waste material, preventing global warming contributions.

Data is the New Doping

อุปกรณ์สวมใส่และเซ็นเซอร์

Hykso Punch Sensor

Punch Velocity



Punch Force



Embedded in boxing gloves. Measures punch speed, force, and type (hook, jab) to expose weaknesses and adjust training without altering the fighter's feel.

Whoop Strap Analytics

Heart Rate Variability



Recovery %



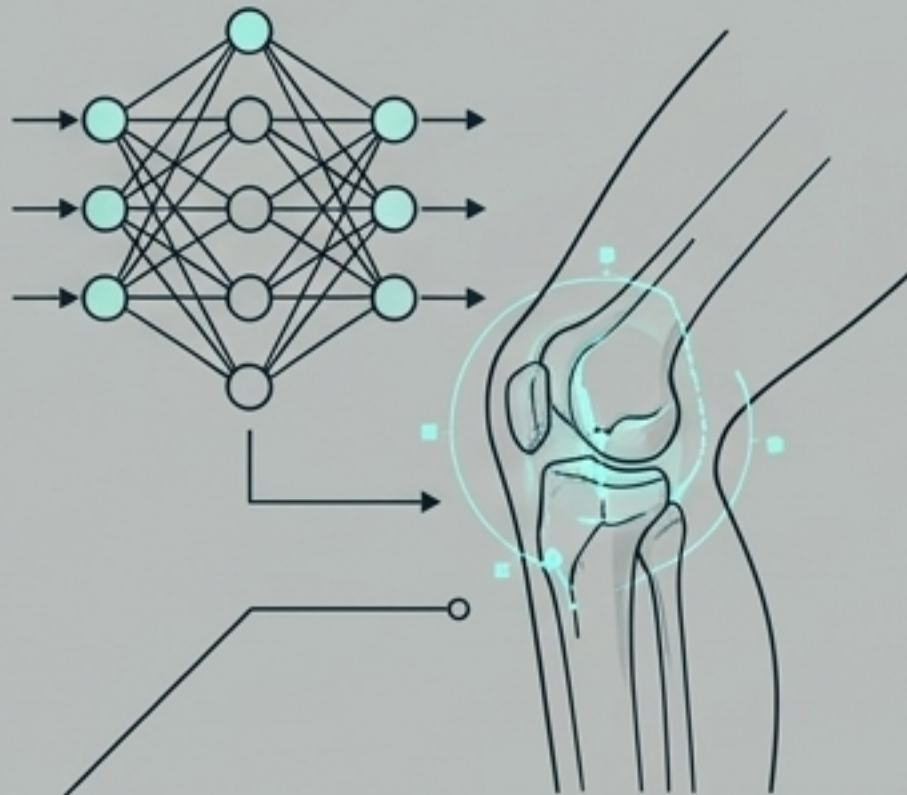
Analyzes past training to predict future physical development and calculates precise daily recovery needs.

Athletes use data for daily micro-adjustments, not just passive tracking.

AI, VR, and The Umpire in the Sky

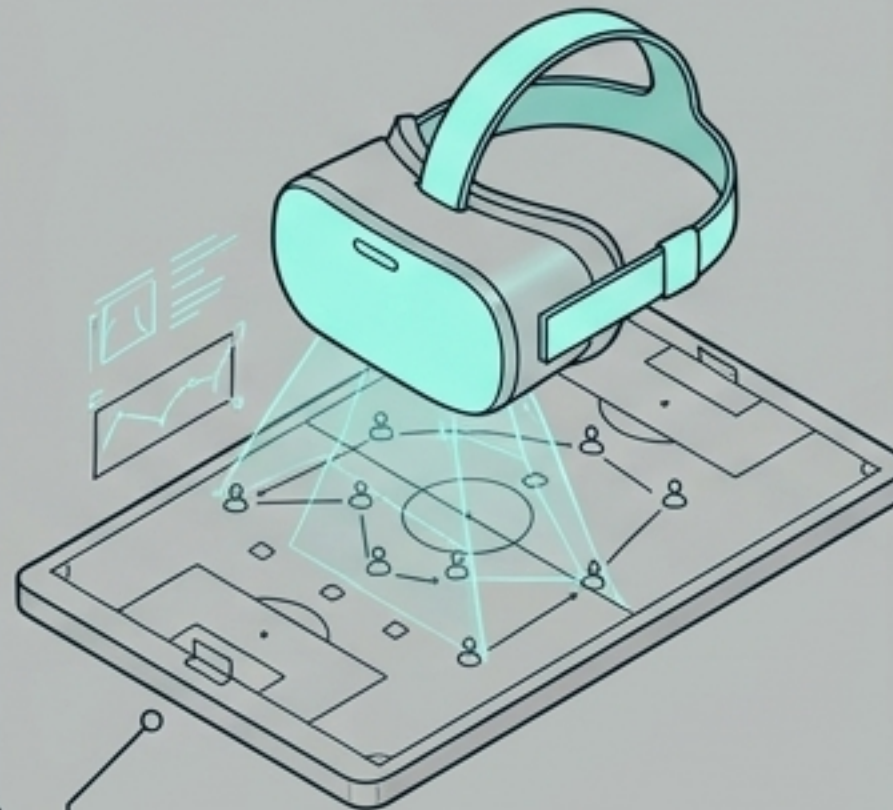
เทคโนโลยีเพื่อความแม่นยำและปลอดภัย

AI Analytics



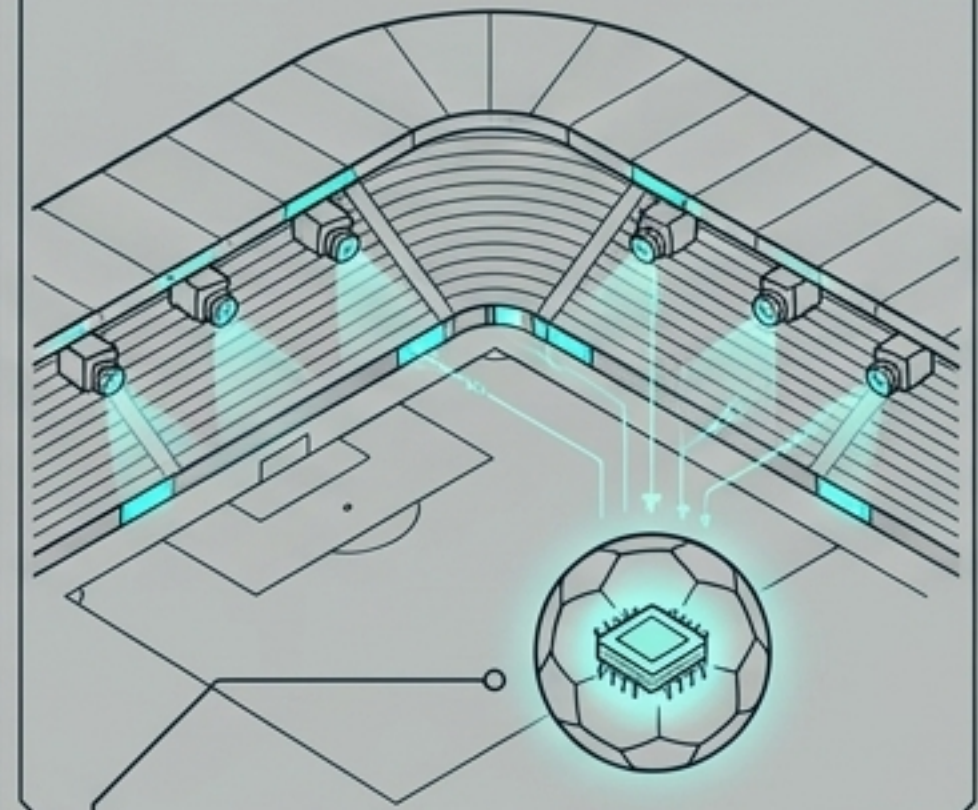
AI ingests sensor data to predict injury risks, prevent overtraining, and customize race plans.

VR Simulation



Creates perfectly safe, simulated field practice, removing the risk of physical impact during drills.

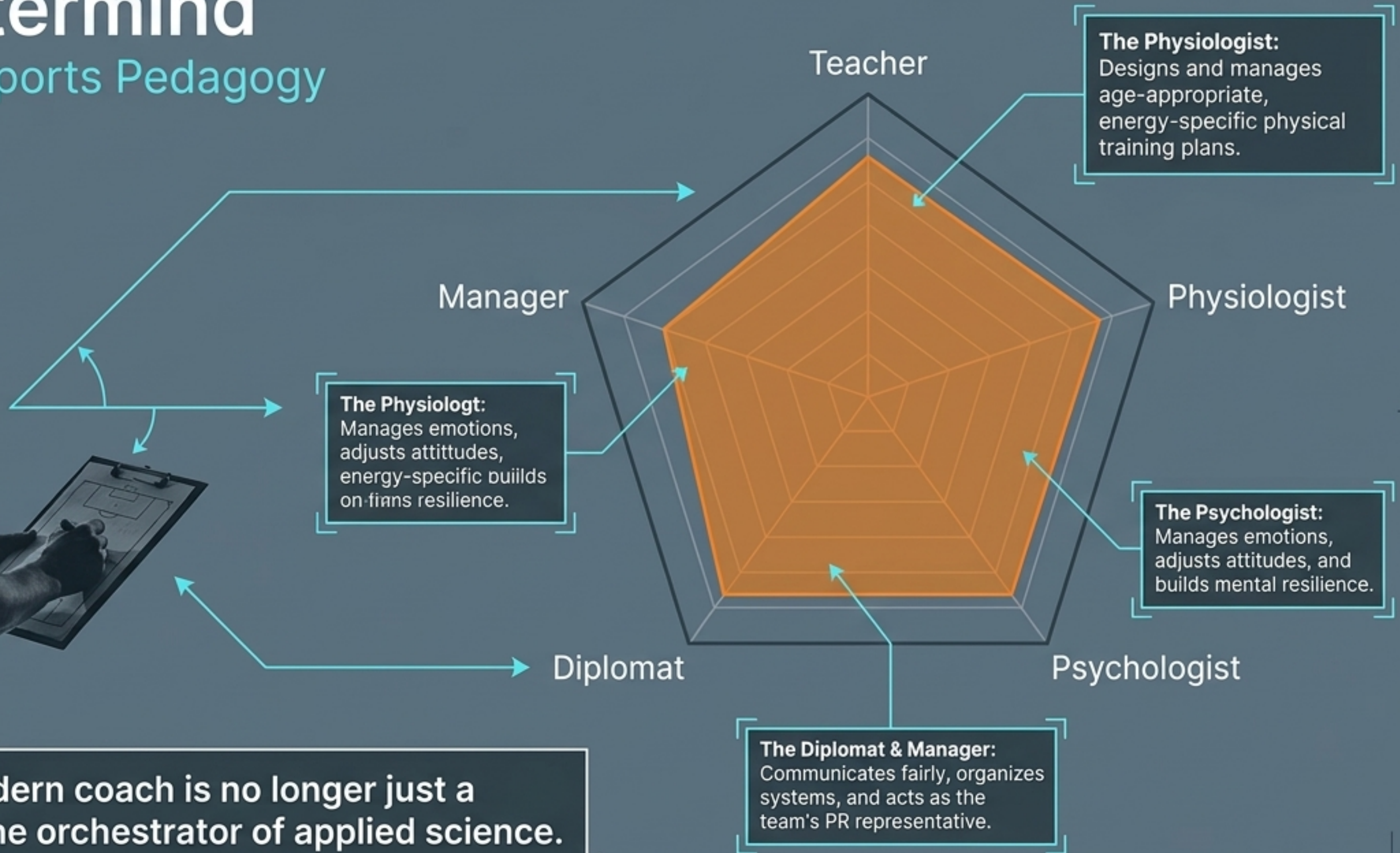
SAOT (FIFA)



Semi-Automated Offside Technology.
12 stadium cameras + internal ball sensors feed real-time offside alerts to human referees.

The Mastermind

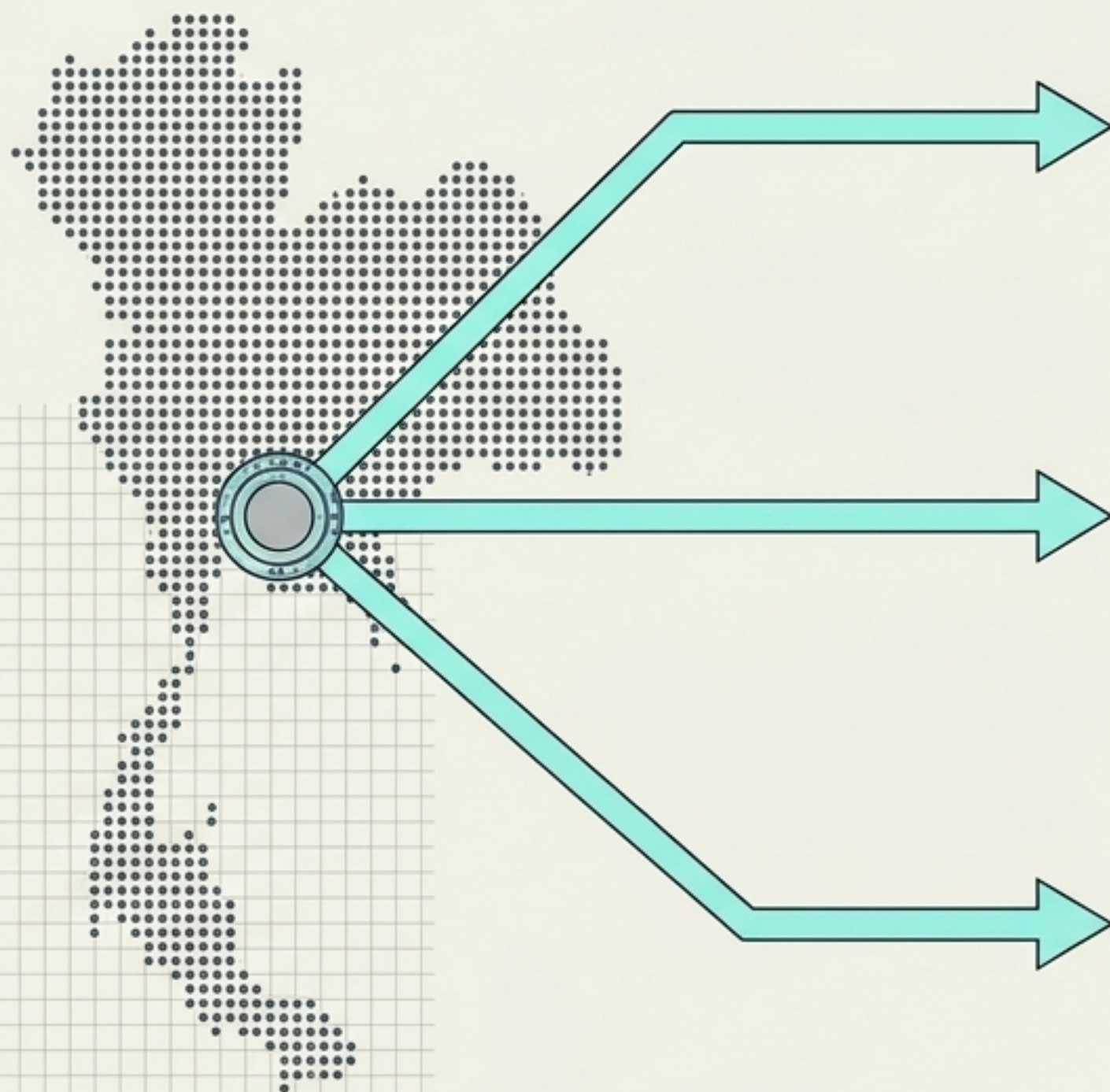
ผู้ฝึกสอนกีฬา - Sports Pedagogy



Summary: The modern coach is no longer just a 'trainer'; they are the orchestrator of applied science.

Driving Thai Sports Forward

พลังขับเคลื่อนกีฬาไทย



The Investment

PTT Group's Soft Power initiative injects **200 Million THB** into **20+** national sports associations (Football, Tennis, Swimming, Boxing, Taekwondo).

The ROI (Real World Impact)

Elevating Thai athletes to the global stage—highlighted by Olympic successes like Panipak (Tennis) winning a 2nd Gold in Taekwondo (Paris 2024), and Janjaem (Bee) securing Boxing Bronze.



The Goal

Inspiring youth, promoting health, and building sustainable national pride.

The Future of Human Performance

บทสรุป: อนาคตของศักยภาพมนุษย์

Unbreakable Will

(The Human Element -
Psychology & Pedagogy)

Biological Optimization

(The Science - Anatomy, Med, Nutrition)

**Peak
Performance**
(ศักยภาพสูงสุด)

Data Precision

(The Tech - AI, Sensors, Gear)

As technology accelerates, human limits are constantly rewritten. What sports technology do you think will break the next world record?
(ร่วมแชร์นวัตกรรมกีฬาที่คุณคิดว่าจะเปลี่ยนโลกในคอมเมนต์!)